

SIH26102

SMART GOVERNANCE

PRODUCTION VERIFIED

ZERO MOCK DATA

# JanDrishti — Architecture & Governance Blueprint

Parliamentary Intelligence, MPLADS Statutory Audit & Separation of Powers

COVERAGE

542 LS + 236 RS  
MPs

RECONCILED ASSETS

102,437 Works

FINANCIAL VARIANCE

₹0.00 (Balanced)

STATUTORY TIERS

6 Governance Ranks



## 1. Full System Architecture & Data Flow (Miro Board)

Decoupled 3-tier architecture with dual-engine persistence and strict policy gating.

### TIER 1: FRONTEND CLIENT LAYER

React 19 + TypeScript + Vite

#### Public Citizen Portal

OverviewPage.tsx,  
MpExplorerPage.tsx National  
geographic maps & MP dossier  
inspection.

#### Role Workspaces

workspaces/\*.tsx (MP,  
District, State) Dedicated  
command hubs for authorized  
personnel.

#### Intelligence Hub

AnomalyCenterPage.tsx,  
CasesAlertsPage.tsx Cartel,  
Benford, and duplication  
investigation UI.

#### API Client & Interceptor

frontend/src/api/client.ts  
JWT Bearer injection, auto-  
retry, and 403 boundary  
handling.

▼ HTTP / REST with HMAC-SHA256 JWT Bearer Tokens ▼

### TIER 2: API GATEWAY & SECURITY ENFORCEMENT

FastAPI + Pydantic v2

#### FastAPI Router Gateway

backend/main.py 95+ REST  
endpoints with CORS and strict  
input schemas.

#### Statutory RBAC & ABAC

backend/rbac\_abac.py Multi-  
rank permission interceptor and  
field immutability.

#### Jurisdiction Scoping

backend/scope.py Mandatory  
SQL territorial clauses  
(National/State/District/MP).

#### Auth Session Engine

backend/auth.py Statutory role  
resolution, demo credentials,  
token signing.

▼ Validated & Jurisdiction-Scoped Service Execution ▼

## TIER 3 & 4: BUSINESS ENGINES & FORENSIC INTELLIGENCE

Python 3.13 Services

### Governance Service

`backend/gov_service.py` MP recommendation lifecycle, sanctions, contractor assign.

### Workflow State Machine

`backend/workflow.py` Statutory transition validator (prevents illegal skips).

### Forensic Intelligence Hub

`backend/intelligence.py` Benford's Law, Levenshtein duplication, Cartel HHI indices.

### Composite Risk Engine

`backend/risk_engine.py`  
Weighted multi-signal risk index and auto-escalation.

## ▼ Dual Database Engine Failover Driver ▼

## TIER 5: PERSISTENCE LAYER (Dual Database Engine)

Supabase PostgreSQL + SQLite 3

### Primary Cloud Database

Supabase PostgreSQL (AWS Tokyo) PostgREST API, RLS security policies, relational ledger.

### Resilient Local Fallback

`database/mplads.db` (SQLite 3) Zero-config offline mode with busy-timeout pragmas.

### Unified DB Engine

`backend/db_engine.py`  
Transparent cursor wrapper adapting queries for both engines.

### Chained Audit Trail

`backend/audit_logger.py`  
Immutable log of all user actions, state changes, and client IPs.

## 2. Statutory Authority Hierarchy & Separation of Powers

Strict compliance with MoSPI operational guidelines across Indian public finance tiers.

### **RANK 1** **MINISTRY\_ADMIN** Central MoSPI

National supervisory power. Can view all Indian states, configure system parameters, freeze suspicious dockets, and handle parliamentary questions.

**Jurisdiction:** Pan-India (National)  
**Workspaces:** `/workspaces/ministry`

### **RANK 2** **STATE\_NODAL\_AUTHORITY** State Planning Dept

Reviews technical approvals from district collectors, manages state allocation quotas, grants statutory financial sanctions, or rejects non-compliant works.

**Jurisdiction:** Designated State  
( `user.state` )  
**Workspaces:** `/workspaces/state`

### **RANK 3** **DISTRICT\_AUTHORITY** District Collector / DM

Performs techno-economic scrutiny on MP recommendations, assigns implementing agencies/contractors, records field milestones, and uploads geo-tagged photos.

**Jurisdiction:** Designated District  
( `user.district` )  
**Workspaces:** `/workspaces/district`

### **RANK 4 MEMBER OF** **PARLIAMENT** Lok Sabha / Rajya Sabha

Drafts and submits constituency work recommendations within their annual entitlement (₹5 Crore). Tracks sanction velocity and project completion milestones.

**Jurisdiction:** MP Seat ( `user.mp_id` )  
**Workspaces:** `/workspaces/mp`

### **RANK 5 STATUTORY** **AUDITOR** CAG / Forensic Auditor

Independent forensic oversight. Inspects statistical anomalies, cartel formation, and Benford flags. Creates formal audit cases and flags evidence.

**Jurisdiction:** National Independent Oversight  
**Workspaces:** `/workspaces/auditor`

### **RANK 6 CITIZEN** General Public

Open civic transparency. Can search any MP's performance, inspect local village works on interactive maps, and file ground discrepancy feedback.

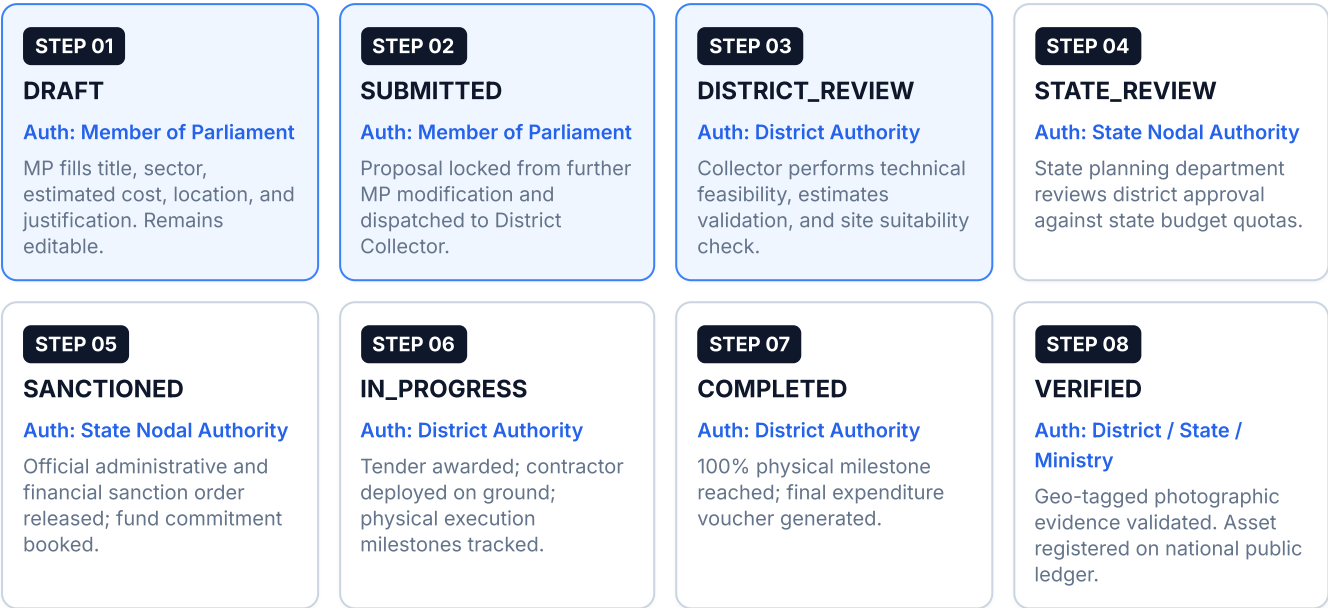
**Jurisdiction:** Read-Only Public Corpus  
**Workspaces:** `/workspaces/citizen`

### **Statutory Financial Immutability Rule (ABAC Enforcement)**

Fields such as `sanctioned_amount`, `expenditure_amount`, `work_id`, and `internal_transaction_id` strictly reject generic HTTP PUT/PATCH requests. Financial mutations can only occur via reconciled Treasury Voucher ledger operations with dual-signoff.

### 3. Statutory Recommendation Lifecycle State Machine

Guaranteed forward-only transitions defined in `backend/workflow.py`.



#### Return for Correction & Rejection Paths

If a proposal contains cost defects, the District Authority transitions it to `RETURNED_FOR_CORRECTION`, re-opening MP edit access. If a proposal falls under the MoSPI Prohibited Works List, it transitions to `REJECTED` (Terminal State).

### 4. Pre-Disbursement Forensic Audit Intelligence

Real-time statistical anomaly screening across 6 autonomous detection engines.

| Forensic Engine           | Algorithmic Principle                                                                                                           | Default Weight   | Threshold Trigger                |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------------|
| 1. Semantic Duplication   | Levenshtein distance and Jaccard similarity across project titles, sectors, and GPS locations to flag duplicate work funding.   | 0.35 (35%)       | Similarity index > 0.85          |
| 2. Vendor Cartelization   | Herfindahl-Hirschman Index (HHI) calculating vendor revenue concentration and repeat contract capture in a single constituency. | 0.25 (25%)       | HHI score > 2,500 (Monopolistic) |
| 3. Benford's Law Analysis | Chi-square and log-frequency goodness-of-fit test on the leading digits of disbursement voucher amounts.                        | 0.20 (20%)       | p-value < 0.05 deviation         |
| 4. Expenditure Velocity   | Z-score standard deviation anomaly detector flagging uncharacteristic "March-rush" spending bursts before fiscal close.         | 0.20 (20%)       | Z-score > 3.0 standard dev       |
| 5. Predictive Delay Model | Random Forest regression estimating expected completion timelines based on historical district performance.                     | Auxiliary Signal | Predicted delay > 180 days       |

## ⚡ 5. Step-by-Step Code Execution Trace (Front-to-Back)

How a user request executes across all system layers.

### 1 User Interaction in Frontend UI

An MP completes proposal details in `frontend/src/pages/workspaces/MpWorkspace.tsx` and clicks "Submit Recommendation".

### 2 HTTP Request Dispatch with JWT

`frontend/src/api/client.ts` attaches the stored session JWT to `POST /api/v1/recommendations/submit`.

### 3 Authentication & Statutory Identity Resolution

FastAPI invokes `backend/auth.py::verify_bearer_token()`, unpacking the token into an `AuthenticatedUser` object with role, state, and mp\_id.

### 4 ABAC & Jurisdiction Boundary Interception

`backend/rbac_abac.py::check_permission()` verifies role privileges, checks field immutability, and ensures the MP only acts within their constituency boundary.

### 5 Workflow State Machine Transition Check

`backend/workflow.py::validate_workflow_transition()` verifies the current status is `DRAFT` and target is `SUBMITTED`, authorized only for `ROLE_MP`.

### 6 Database Mutation & Dual Engine Failover

`backend/gov_service.py` executes the SQL UPDATE via `backend/db_engine.py`, targeting Supabase PostgreSQL with transparent SQLite fallback.

### 7 Cryptographic Chained Audit Logging

`backend/audit_logger.py::record_audit_log()` appends a tamper-proof audit record containing timestamp, user\_id, action, client IP, and state transition.

### 8 Response Serialization & UI Cache Invalidation

FastAPI serializes the updated record via Pydantic v2 schemas; the React client invalidates its cache and renders a success notification.

## 📁 6. Codebase Directory & Module Responsibility Matrix

Primary source code files, classes, and their operational responsibilities.

| File Path                         | Key Functions / Classes                                             | Operational Responsibility                                      |
|-----------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------|
| <code>backend/main.py</code>      | <code>app</code> , 95+ endpoint routes, CORS middleware             | HTTP gateway, request dispatching, error mapping.               |
| <code>backend/rbac_abac.py</code> | <code>check_permission()</code> , <code>ROLE_PERMISSIONS</code>     | Statutory RBAC matrix, ABAC boundary & immutability gating.     |
| <code>backend/workflow.py</code>  | <code>validate_workflow_transition()</code>                         | State machine definitions and transition authority enforcement. |
| <code>backend/scope.py</code>     | <code>jurisdiction_clause()</code> , <code>can_edit_record()</code> | Generates mandatory SQL filters for user territorial scopes.    |

| File Path                                   | Key Functions / Classes                                                                    | Operational Responsibility                                          |
|---------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <code>backend/gov_service.py</code>         | <code>create_recommendation()</code> ,<br><code>review_recommendation()</code>             | Core business logic for recommendations, reviews, and sanctions.    |
| <code>backend/intelligence.py</code>        | <code>intelligence_service</code> , Benford,<br>Dupe, Cartel HHI                           | Statistical intelligence algorithms and forensic signal generation. |
| <code>backend/risk_engine.py</code>         | <code>risk_engine</code> ,<br><code>calculate_composite_risk()</code>                      | Weighted multi-signal risk scoring and audit case queuing.          |
| <code>backend/db_engine.py</code>           | <code>PostgresConnection</code> ,<br><code>get_db_connection()</code>                      | Dual-engine database abstraction for Supabase & SQLite.             |
| <code>frontend/src/pages/workspaces/</code> | <code>MpWorkspace</code> , <code>DistrictWorkspace</code> ,<br><code>StateWorkspace</code> | Role-specific interactive governance user interfaces.               |

## 7. How to Customize & Extend Later

Quick guide for extending the platform with new roles or workflow rules.

### 1. Adding a New Role

1. Define constant in `backend/rbac_abac.py` .
2. Assign rank in `ROLE_HIERARCHY_RANK` .
3. Define action sets in `ROLE_PERMISSIONS` .
4. Add UI page in `frontend/src/pages/workspaces/` .

### 2. Modifying Workflow Rules

1. Edit `RECOMMENDATION_TRANSITIONS` in `backend/workflow.py` .
2. Map authorized roles in `TRANSITION_AUTHORITY` .
3. Add new status badge colors in `frontend/src/utils/formatters.ts` .

### 3. Tweaking Anomaly Weights

1. Adjust `DEFAULT_WEIGHTS` in `backend/risk_engine.py` .
2. Or use live REST API: `PUT /api/v1/risk-weights` to update weights dynamically without redeploying.